

2.

$$I = \int_0^1 [(y')^2 + 12xy] dx$$

$$\frac{dy}{dx} = u$$

$$\frac{d^2y}{dx^2} = \frac{du}{dx}$$

$$\frac{\partial F}{\partial y} - \frac{d}{dx} \left(\frac{\partial F}{\partial y'} \right) = 0 \quad \text{donde } F = (y')^2 + 12xy$$

$$\frac{\partial}{\partial y} [(y')^2 + 12xy] - \frac{d}{dx} \left[\frac{\partial}{\partial y'} ((y')^2 + 12xy) \right] = 0$$

$$12x - \frac{d}{dx} [2y'] = 0 \Rightarrow 12x - \frac{d}{dx} \left[2 \frac{dy}{dx} \right] = 0$$

$$12x - 2y'' = 0 \Rightarrow y'' = \frac{-12x}{-2} = 6x \Rightarrow \frac{d^2y}{dx^2} = 6x \Rightarrow \frac{dy}{dx} = 6x$$

$$\int du = \int 6x dx \Rightarrow u = 3x^2 \Rightarrow \frac{dy}{dx} = 3x^2 \Rightarrow \int dy = \int 3x^2 dx \Rightarrow y(x) = x^3$$

$$y'(x) = 3x^2$$

$$I = \int_0^1 [(3x^2)^2 + 12x(x^3)] dx$$

$$= \int_0^1 [9x^4 + 12x^4] dx = \left[9 \frac{x^5}{5} + 12 \frac{x^5}{5} \right]_0^1$$

$$= \frac{9}{5} + \frac{12}{5} = \frac{21}{5}$$

$$3. \quad x^2 + y^2 - z^2 = 0 \quad P_1 = (a, 0, 0) ; P_2 = (-a, 0, \pi)$$

$$x^2 + y^2 = z^2$$

$$(r \cos \varphi)^2 + (r \sin \varphi)^2 = z^2$$

$$r^2 = z^2 \Rightarrow r = z \rightarrow \text{cte}$$

$$\begin{cases} x = a \cos \varphi \\ y = a \sin \varphi \\ z = z \end{cases} \quad \begin{cases} ds^2 = dr^2 + r^2 d\varphi^2 + dz^2 \\ ds^2 = a^2 d\varphi^2 + dz^2 \\ ds = \sqrt{a^2 d\varphi^2 + dz^2} = \sqrt{a^2 + \left(\frac{dz}{d\varphi}\right)^2} d\varphi \end{cases}$$

$$\Rightarrow \int_0^\pi \sqrt{a^2 + \left(\frac{dz}{d\varphi}\right)^2} d\varphi = \int_0^\pi \sqrt{a^2 + (z')^2} d\varphi$$

$$L = \sqrt{a^2 + (z')^2}$$

$$\frac{\partial L}{\partial z} = 0 ; \quad \frac{\partial L}{\partial z'} = \frac{2z'}{2(a^2 + (z')^2)^{1/2}} = \frac{z'}{(a^2 + (z')^2)^{1/2}}$$

$$\frac{\partial}{\partial \varphi} \left(\frac{\partial L}{\partial z'} \right) - \frac{\partial L}{\partial z} = 0 \Rightarrow \frac{\partial}{\partial \varphi} \left(\frac{z'}{(a^2 + (z')^2)^{1/2}} \right) = 0$$

$$\frac{z'}{(a^2 + (z')^2)^{1/2}} = k \Rightarrow z(\varphi) = k\varphi + c$$

$$\text{Para } z(0) = 0$$

$$z(0) = k(0) + c = 0 \\ c = 0$$

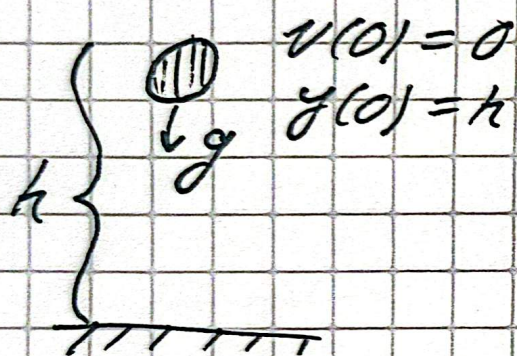
$$\text{Para } z(\pi) = \pi$$

$$z(\pi) = k\pi + 0 = \pi \\ k = 1$$

$$z(\varphi) = \varphi$$

$$\begin{cases} x = a \cos(\varphi) \\ y = a \sin(\varphi) \\ z = \varphi \end{cases}$$

4.



$$L = \frac{m}{2} \dot{y}^2 - mgy$$

$$\frac{dL}{d\dot{y}} = m\dot{y} \rightarrow \frac{d}{dt}(m\dot{y}) = m\ddot{y}$$

$$\frac{dL}{dy} = -mg \rightarrow m\ddot{y} + mg = 0 \rightarrow \ddot{y} = -g$$

$$\frac{d}{dt}\left(\frac{dy}{dt}\right) = -g \rightarrow \int d\left(\frac{dy}{dt}\right) = -g \int dt \rightarrow \frac{dy}{dt} = -gt + A$$

$$\dot{y}(0) = 0 \rightarrow 0 = A \rightarrow \int dy = -g \int t dt \rightarrow y = -\frac{g}{2} t^2 + B$$

$$y(0) = h = B \rightarrow y = h - \frac{g}{2} t^2$$

5

$$I = \frac{m^2 \dot{x}^4}{12} + m \dot{x}^2 f(x) - f^2(x)$$

$$\frac{\partial I}{\partial x} = m \dot{x}^2 f'(x) - 2f(x)f'(x); \quad \frac{\partial I}{\partial \dot{x}} = \frac{m^2 \dot{x}^3}{3} + 2m f(x) \dot{x}$$

$$\frac{d}{dt} \left(\frac{m^2 \dot{x}^3}{3} + 2m f(x) \dot{x} \right) - (m \dot{x}^2 - 2f(x)f'(x)) = 0$$

$$m^2 \dot{x}^2 \ddot{x} + 2m \dot{x} f'(x) \dot{x} + 2m f(x) \ddot{x} - m \dot{x}^2 f'(x) + 2f(x)f'(x) = 0$$

$$m^2 \dot{x}^2 \ddot{x} + 2m \dot{x}^2 f'(x) + 2m \ddot{x} f(x) - m \dot{x}^2 f'(x) + 2f(x)f'(x) = 0$$

$$m^2 \dot{x}^2 \ddot{x} + \dot{x}^2 (2m f'(x) - m f'(x)) + 2m \ddot{x} f(x) + 2f(x)f'(x) = 0$$

$$m^2 \dot{x}^2 \ddot{x} + m \dot{x}^2 f'(x) + 2m \ddot{x} f(x) + 2f(x)f'(x) = 0$$

$$\ddot{x} (m^2 \dot{x}^2 + 2m f(x)) + f'(x) (m \dot{x}^2 + 2f(x)) = 0$$

$$\ddot{x} (m^2 \dot{x}^2 + 2m f(x)) = -f'(x) (m \dot{x}^2 + 2f(x))$$

$$m \ddot{x} (m \dot{x}^2 + 2f(x)) = -f'(x) (m \dot{x}^2 + 2f(x))$$

$$m \ddot{x} = -f'(x) \Rightarrow m \dot{x} + f(x) = 0$$

Sea $u = \frac{dx}{dt} = \dot{x}$

$$\ddot{x} = \frac{du}{dt} = \frac{du}{dx} \frac{dx}{dt} = u \frac{du}{dx}$$

entonces

$$m \left(u \frac{du}{dx} \right) + f'(x) = 0$$

$$u du = -\frac{1}{m} f'(x) dx \Rightarrow \int u du = -\frac{1}{m} \int f'(x) dx$$

$$\frac{u^2}{2} = -\frac{f(x)}{m} + C$$

$$u^2 = -\frac{2f(x)}{m} + C \Rightarrow \dot{x}^2 = C_1 - \frac{2f(x)}{m} \Rightarrow \frac{dx}{dt} = \sqrt{C_1 - \frac{2f(x)}{m}}$$

$$\int \frac{dx}{\sqrt{C_1 - \frac{2f(x)}{m}}} = \int dt$$